Lesson 2.2 Guided Notes

Session 1 · Building a Histogram

I can group data into equal intervals and build a histogram from a frequency table. · 6.SP.B.4

Word / Spanish	Meaning — complete the blank	Example
histogram / histograma	A graph of _____ bars; each bar counts the values in one interval.	bars over 0–9, 10–19, 20–29
interval / intervalo	A group of numbers, like 10–19. All intervals have the same _____.	10–19 holds 10 whole numbers
frequency / frecuencia	_____ values land in one interval.	10–19: 3 values

Key idea

Intervals must be the same _____ and must not _____.

The bar heights add up to the _____ number of values.

Worked example

Walk times (min): 4, 15, 9, 22, 11, 18, 27. Use 0–9, 10–19, 20–29.

0–9 holds 4 and 9, so its frequency is _____.

10–19 holds 15, 11 and 18: frequency _____.

20–29 holds 22 and 27: frequency _____.

Answer: Check: 2 + 3 + 2 = _____ values

You try

1. Ages: 13, 27, 21, 8, 16, 24, 29, 18. Find each frequency.
a. 0–9: _____
b. 10–19: _____
c. 20–29: _____

_____

2. A histogram shows 10 values. Two bars have heights 4 and 2. How tall is the third bar?

_____

Sum it up: To build a histogram, I first ___, then ___.

_____

Session 2 · Describing Histogram Shape

I can describe the shape of a histogram using its peak, spread, and balance. · 6.SP.B.4

Word / Spanish	Meaning — complete the blank	Example
peak / pico	The interval with the _____ bar.	bars 2, 7, 3: peak is the middle bar
symmetric / simétrico	The left and right sides _____.	1, 4, 8, 4, 1
skewed right / sesgado a la derecha	The long _____ stretches to the right.	9, 5, 2, 1, 1
skewed left / sesgado a la izquierda	The long tail stretches to the _____.	1, 2, 3, 6, 8

Key idea

The _____ shows where a typical value falls.

Name a skewed shape for its _____, not for where the tall bars are.

Worked example

Bars of height 1, 3, 6, 9 sit over 0–9, 10–19, 20–29, 30–39. Describe the shape.

The tallest bar, 9, is over _____, so that is the peak.

The bars shrink toward the _____, so the tail is on the left.

Answer: The shape is _____.

You try

1. Bar heights from left to right. Name each shape.
a. 2, 5, 9, 5, 2 _____
b. 10, 6, 3, 2, 1 _____

_____

2. Bars of 3, 8, 4 sit over 60–69, 70–79, 80–89. Name the peak and the number of values.

_____

Sum it up: A histogram is skewed right when ___ because ___.

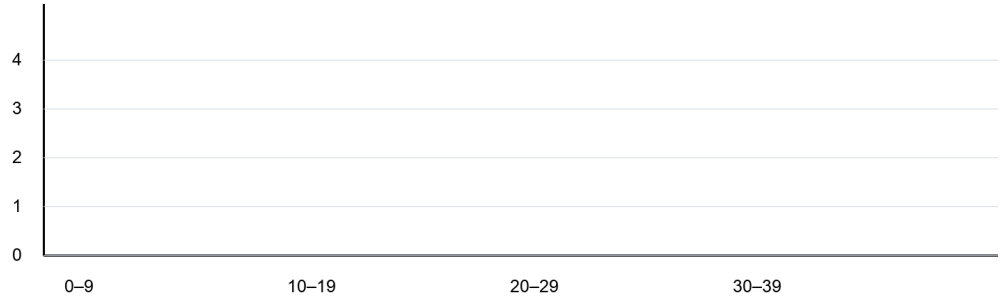
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Session 1 · Model, reason, and explain

Building a Histogram · 6.SP.B.4

Success: represent the quantities, justify the relationship, and check the result with the correct units.

Reading times are 0, 9, 10, 19, 20, 24, 29, 30 minutes. Use intervals 0–9, 10–19, 20–29, 30–39. Make a frequency table and histogram. Explain where 10 and 20 belong.



_____

_____

_____

Extend your reasoning

For measurements that include decimals, rewrite the intervals so 9.5 also has exactly one home.

_____

_____

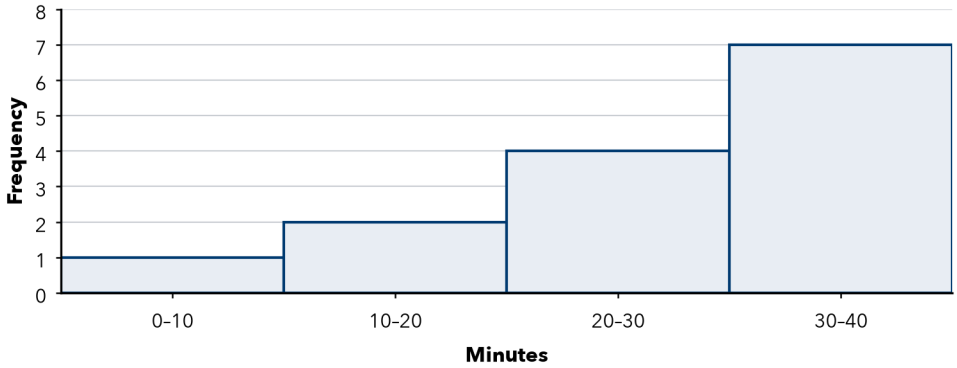
Self-check: Did my model, calculation, explanation, and units agree?

Session 2 · Model, reason, and explain

Describing Histogram Shape · 6.SP.B.4

Success: represent the quantities, justify the relationship, and check the result with the correct units.

The histogram has interval frequencies 1, 2, 4, 7 from left to right. Which side has the tail? Explain why the location of the tallest bar does not name the tail.



_____

_____

_____

Extend your reasoning

Design four interval frequencies that make a symmetric histogram with eight observations.

_____

_____

Self-check: Did my model, calculation, explanation, and units agree?